

QUESTION 20

Develop a Document Question Answering System Using Retrieval-Augmented Generation (RAG)

Core Architecture: Retrieval + Context + Answer Generation

Retriever: FAISS | Generator: google/flan-t5-small

1. Aim

To develop a document-based Question Answering system using Retrieval-Augmented Generation (RAG), where relevant document information is retrieved from a vector database and supplied as context to an answer-generation model.

2. Problem Statement

A language model may not have direct access to information contained in a document collection. A RAG system addresses this by retrieving relevant information first and generating an answer from the retrieved context.

3. Objectives

- Create a document knowledge base.
- Generate vector embeddings for the documents.
- Store document vectors in a FAISS retrieval index.
- Convert the user question into an embedding.
- Retrieve the most relevant documents.
- Construct context from retrieved documents.
- Generate a natural-language answer using FLAN-T5.

4. RAG Architecture

Retrieval-Augmented Generation (RAG) Architecture

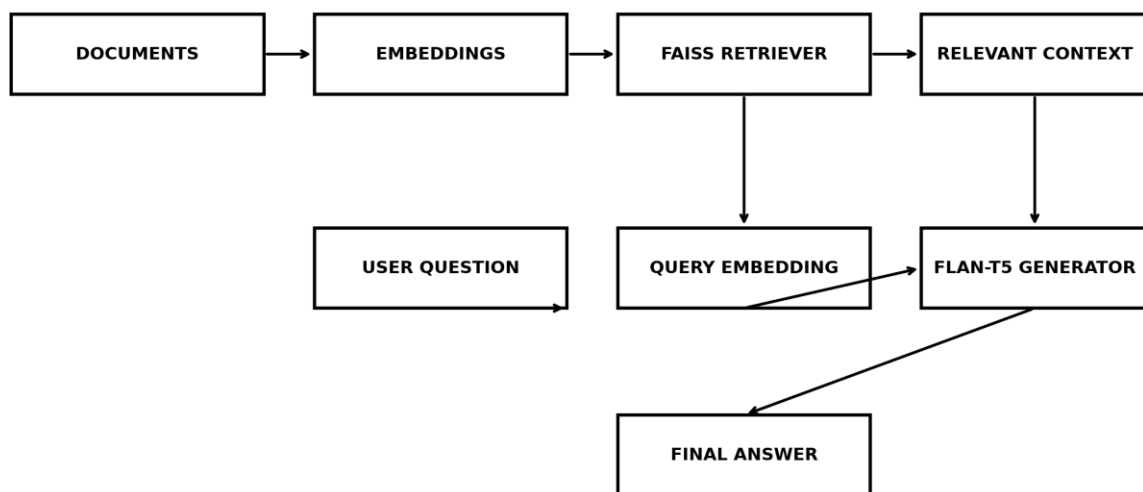


Figure 1: Retrieval-Augmented Generation architecture.

5. Working Principle

The system performs retrieval followed by generation. Documents and the question are converted into embeddings. FAISS retrieves the most relevant documents. The retrieved text is combined into context and passed with the question to FLAN-T5, which generates the final answer.

6. RAG Pipeline

1. Load embedding model.
2. Prepare document knowledge base.
3. Generate document embeddings.
4. Create FAISS vector index.
5. Accept user question.
6. Generate question embedding.
7. Retrieve top relevant documents.
8. Construct context.
9. Generate answer with FLAN-T5.
10. Display answer and retrieval information.

7. Technology Stack

Component	Implementation
Programming Language	Python
Embedding Model	all-MiniLM-L6-v2
Embedding Dimension	384
Vector Database	FAISS
Index	IndexFlatIP
Generator	google/flan-t5-small
Retrieved Documents	2

8. Python Implementation

```
embedding_model = SentenceTransformer("all-MiniLM-L6-v2")
document_embeddings = embedding_model.encode(
    documents, convert_to_numpy=True
).astype("float32")
faiss.normalize_L2(document_embeddings)

index = faiss.IndexFlatIP(document_embeddings.shape[1])
index.add(document_embeddings)

question_embedding = embedding_model.encode(
    [question], convert_to_numpy=True
```

```

).astype("float32")
faiss.normalize_L2(question_embedding)

scores, indices = index.search(question_embedding, 2)

context = "\n\n".join(
    documents[int(i)] for i in indices[0]
)

prompt = f"""Answer the question using only the context.

Context:
{context}

Question:
{question}

Answer:"""

inputs = tokenizer(prompt, return_tensors="pt",
                    truncation=True, max_length=512)
outputs = answer_model.generate(
    **inputs, max_new_tokens=80, do_sample=False
)
answer = tokenizer.decode(
    outputs[0], skip_special_tokens=True
)

```

9. Test Question

What is machine learning?

10. Actual Implementation Output

```
DOCUMENT QUESTION ANSWERING USING RAG

Question:
What is machine learning?

RETRIEVED DOCUMENT 1
Similarity Score : 0.8383
Machine learning is a branch of artificial intelligence
that enables computers to learn patterns from data.

RETRIEVED DOCUMENT 2
Similarity Score : 0.5760
Deep learning is a subset of machine learning that uses
artificial neural networks with multiple layers.

GENERATED ANSWER
a branch of artificial intelligence that enables computers
to learn patterns from data

RAG PIPELINE INFORMATION
Embedding Model      : all-MiniLM-L6-v2
Embedding Dimension  : 384
Vector Database      : FAISS
Index Type           : IndexFlatIP
Stored Documents     : 5
Retrieved Documents  : 2
Generation Model      : google/flan-t5-small

RAG QUESTION ANSWERING COMPLETED SUCCESSFULLY
```

Figure 2: Actual RAG implementation output.

11. Retrieved Context Results

Rank	Similarity Score	Retrieved Context
1	0.8383	Machine learning is a branch of artificial intelligence that enables computers to learn patterns from data. It can be used for prediction, classification, recommendation, and decision-making.
2	0.5760	Deep learning is a subset of machine learning that uses artificial neural networks with multiple layers to learn complex patterns from large amounts of data.

12. Retrieval Score Analysis

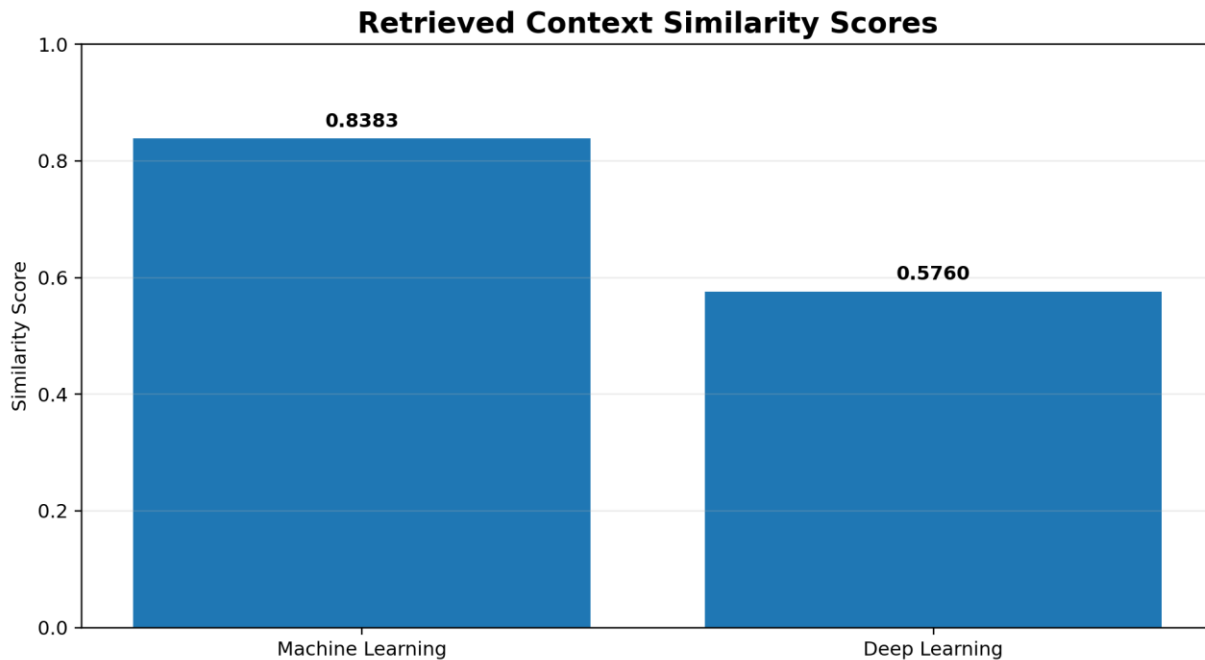


Figure 3: Similarity scores of retrieved documents.

The machine learning document achieved the highest similarity score of 0.8383, making it the primary source of context. The deep learning document was second with 0.5760. Both retrieved documents were supplied to the answer-generation model.

13. Generated Answer Analysis

For the question “What is machine learning?”, the generator produced: “a branch of artificial intelligence that enables computers to learn patterns from data”. This answer is directly supported by the highest-ranked retrieved document.

14. Result

The document Question Answering system successfully retrieved two relevant documents using FAISS and generated an answer using FLAN-T5. The highest retrieval score was 0.8383, and the generated answer correctly described machine learning.

15. Advantages

- Uses external document knowledge during answer generation.
- Separates retrieval from answer generation.
- Can be updated by changing the document knowledge base.
- Provides a foundation for document-based AI assistants.
- Supports retrieval-grounded answers.

16. Applications

- Document question answering
- Knowledge-base assistants
- Educational AI systems
- Technical support systems
- Enterprise document assistants
- RAG-based chatbots

17. Conclusion

A document Question Answering system using Retrieval-Augmented Generation was successfully implemented. The system generated document embeddings, stored them in FAISS, retrieved the top two relevant documents, constructed context, and generated an answer using google/flan-t5-small. The machine learning document achieved 0.8383 similarity and supplied the key information used in the final answer.